

Super-Eddington Little Blue Dots Over-Reionize Helium

JWST's faint AGN may complete helium reionization too early unless only a small fraction are super-Eddington.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint active galaxies at cosmic dawn, dubbed Little Blue Dots. Their broad lines and blue continua suggest super-Eddington accretion with strong extreme-UV but weak X-rays.

Did these super-Eddington black holes reionize helium too early, conflicting with observations?

What this paper is selling

The authors test whether super-Eddington Little Blue Dots can drive helium reionization. They find that if most are super-Eddington, helium reionizes too early, violating current constraints.

Key results to remember

- Helium reionization ending after $z \approx 3$ limits super-Eddington Little Blue Dots to $\lesssim 10\%$ of the population.
- A majority super-Eddington scenario requires tiny escape fractions ($\lesssim 1.5\%$) or lower abundance.
- Constraints depend on extreme-UV spectra of super-Eddington black holes, motivating further study.

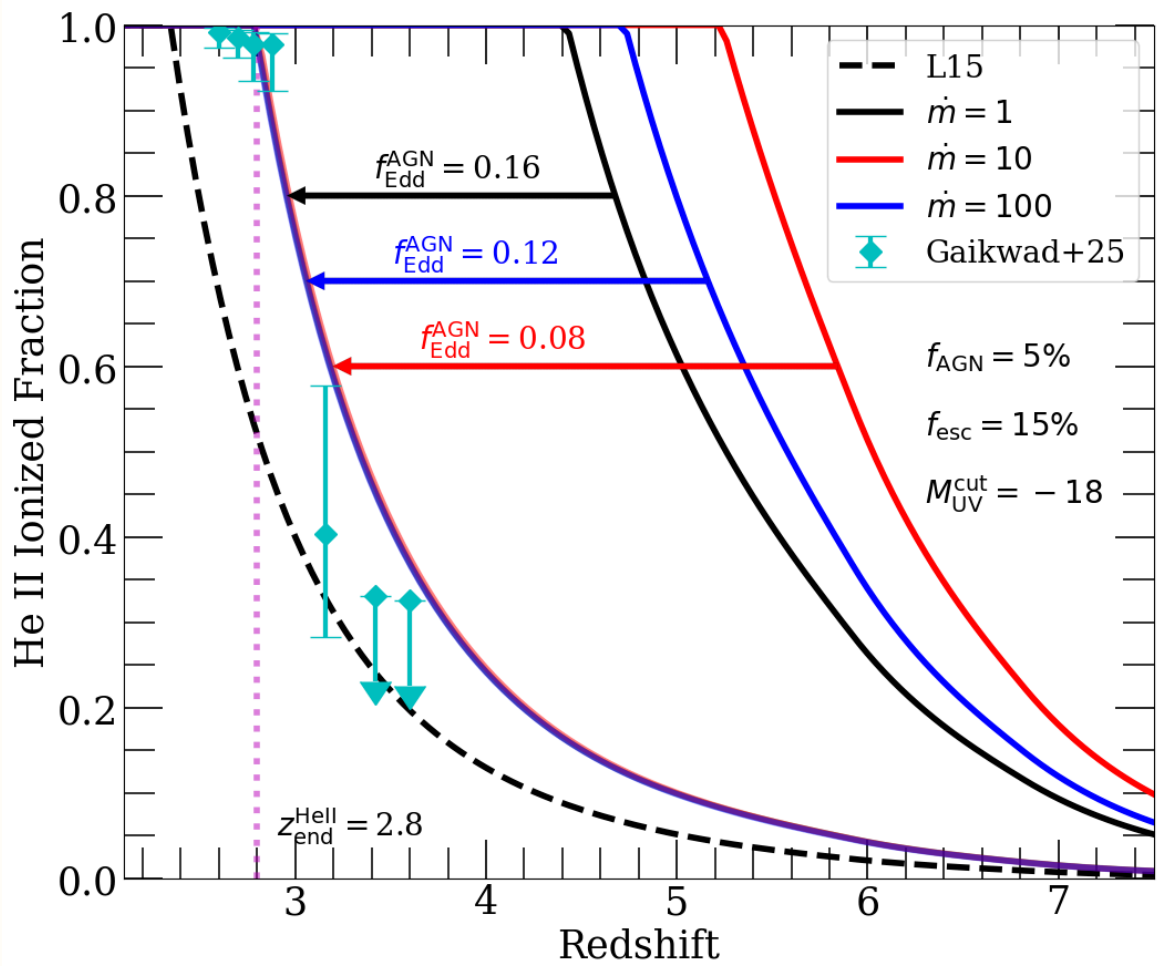


Figure 3 | Helium reionization histories for different super-Eddington spectra, showing that completing reionization by $z=2.8$ requires only a small fraction of AGN.

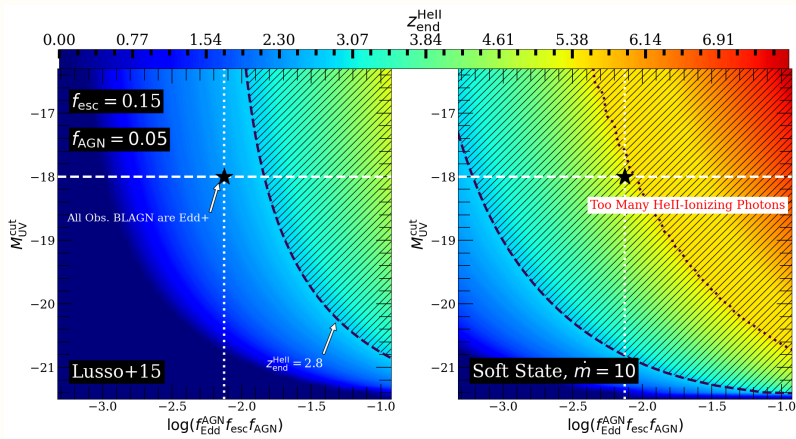


Figure 4 | Excluded parameter space from helium reionization constraints, with the hatched region disfavored for super-Eddington Little Blue Dots.